Programming Assignment - Solutions (C++)

Question 1

Write a program takes input number n then display the natural numbers to n and their sum.

Code:

```
#include <iostream>
using namespace std;

int main() {
    int n, sum = 0;
    cin >> n;
    cout << "The natural numbers are: ";
    for (int i = 1; i <= n; i++) {
        cout << i << " ";
        sum += i;
    }
    cout << endl;
    cout << "The Sum of Natural Number upto " << n << " terms : " << sum << endl;
    return 0;
}
```

Question 2

Write a program that takes an integer n as input and displays its multiplication table up to 10. Hint: Use a loop that starts from 1 and ends at 10, multiplying n by the loop counter each time.

Code:

```
#include <iostream>
using namespace std;

int main() {
    int n;
    cin >> n;
    for (int i = 1; i <= 10; i++) {
        cout << n << " x " << i << " = " << n * i << endl;
    }
    return 0;
}
```

Question 3

Write a program that takes the input number n and then display the n terms of odd natural numbers and their sum. Note: When you input 10 the result should be 10 odd numbers like the example.

Code:

```
#include <iostream>
using namespace std;

int main() {
    int n, sum = 0;
    cin >> n;
    cout << "The odd numbers are: ";
    for (int i = 1; i <= n; i++) {
        int odd = 2 * i - 1;
        cout << odd << " ";
        sum += odd;
    }
    cout << endl;
    cout << "The Sum of odd Natural Numbers upto " << n << " terms: " << sum << endl;
    return 0;
}
```

Question 4

Write a program to enter the numbers till the user wants and at the end, it should display the count of positive, negative and zeros entered. Hint: First ask the user how many numbers he wants to check, then make a loop that loops till the number he entered.

Code:

```
#include <iostream>
using namespace std;

int main() {
    int n, x, pos = 0, neg = 0, zero = 0;
    cout << "How many Number you want to check ?" << endl;
    cin >> n;
    cout << "Enter " << n << " numbers" << endl;
    for (int i = 0; i < n; i++) {
        cin >> x;
        if (x > 0) pos++;
        else if (x < 0) neg++;
        else zero++;
    }
    cout << "You Entered " << pos << " postive numbers And "
         << neg << " negative and " << zero << " Zero" << endl;
    return 0;
}
```

Question 5

Write a program to find the factorial value of any number entered through the keyboard. Hint: Factorial 5 = 5! = 5x4x3x2x1, Factorial 8 = 8! = 8x7x6x5x4x3x2x1.

Code:

```
#include <iostream>
using namespace std;

int main() {
    int n;
    long long fact = 1;
    cin >> n;
    for (int i = 1; i <= n; i++) {
        fact *= i;
    }
    cout << fact << endl;
    return 0;
}
```

Question 6

Write a program that takes input from the user as an integer and then outputs the number with the digits reversed. For example, if the input is 12345, the output should be 54321. Hint: dividing an integer by 10 removes the first digit from the right (1523/10 = 152). Hint 2: the remainder over 10 gives the first digit from the right (1523%10 = 3).

Code:

```
#include <iostream>
using namespace std;

int main() {
    int n, rev = 0;
    cin >> n;
    while (n != 0) {
        int digit = n % 10;     // last digit
        rev = rev * 10 + digit; // add it to the reversed number
        n = n / 10;             // remove last digit
    }
    cout << rev << endl;
    return 0;
}
```

Question 7

Write a program that displays all the numbers from 100 to 1,000, that are divisible by 5 and 6. Numbers are separated by exactly one space. Hint: a number is divisible by 5 if number%5=0 and divisible by 6 if number%6=0.

Code:

```
#include <iostream>
using namespace std;

int main() {
    bool first = true;
    for (int i = 100; i <= 1000; i++) {
        if (i % 5 == 0 && i % 6 == 0) {
            if (!first) cout << " ";
            cout << i;
            first = false;
        }
    }
    cout << endl;
    return 0;
}
```

Question 8

Write a program that takes integer as input and print yes if number is prime number else print no. Hint: a prime number is a number that is divisible by itself and 1 only.

Code:

```
#include <iostream>
using namespace std;

int main() {
    int n;
    bool prime = true;
    cin >> n;
    if (n < 2) prime = false;
    for (int i = 2; i < n; i++) {
        if (n % i == 0) {
            prime = false;
            break;
        }
    }
    if (prime) cout << "yes" << endl;
    else cout << "no" << endl;
    return 0;
}
```

Question 9

Write a program to calculate the power of a number without using built-in math functions. Take two integers from the user: the base and the exponent. Hint: you need to use a loop to multiply the base by itself exponent number of times (e.g., 2^3 = 2 x 2 x 2).

Code:

```
#include <iostream>
using namespace std;

int main() {
    int base, exponent;
    long long result = 1;
    cout << "Enter base: ";
    cin >> base;
    cout << "Enter exponent: ";
    cin >> exponent;
    for (int i = 1; i <= exponent; i++) {
        result *= base;
    }
    cout << result << endl;
    return 0;
}
```

Question 10

Write a program to ask the user how many numbers they want to enter. Then, read those numbers and display the largest (maximum) and smallest (minimum) numbers entered. Hint: create variables like max and min. Assume the first number entered is both the max and the min, then compare every subsequent number against them inside the loop.

Code:

```
#include <iostream>
using namespace std;

int main() {
    int n, x, maxNum, minNum;
    cout << "How many numbers?" << endl;
    cin >> n;
    cout << "Enter " << n << " numbers:" << endl;
    cin >> x;
    maxNum = x;   // first number is both max and min
    minNum = x;
    for (int i = 1; i < n; i++) {
        cin >> x;
        if (x > maxNum) maxNum = x;
        if (x < minNum) minNum = x;
    }
    cout << "Max number is " << maxNum << endl;
    cout << "Min number is " << minNum << endl;
    return 0;
}
```